

EDC BLOCK-003 (Derivation Attempt v3):

Can κ_5^2 Be Fixed by σ ?

Derivation attempt / program note; not a results paper

EDC Research Program

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Status: *WORKING* / *OPEN*

1 Executive Summary

Question: Can the 5D gravitational coupling κ_5^2 be determined by the brane tension σ alone, without importing additional scales?

Outcome: **INCONCLUSIVE.**

Dimensional analysis permits $\kappa_5^2 = C \cdot \sigma^{-3/4}$ (in natural units), where C is an undetermined dimensionless constant. This relation is consistent with Israel junction conditions. However, σ alone cannot fix C ; there exists a residual normalization freedom.

Conclusion: σ determines a natural length scale $\ell_\sigma \sim \sigma^{-1/4}$, but κ_5^2 requires an additional input to be fully specified. BLOCK-003 remains OPEN.

2 Setup: The 5D Action

Action [BL]. The total gravitational action is:

$$S = \underbrace{\frac{1}{2\kappa_5^2} \int_{\mathcal{M}^5} d^5 X \sqrt{-g} (R^{(5)} - 2\Lambda_5)}_{S_{\text{bulk}}} + \underbrace{\frac{1}{\kappa_5^2} \int_{\partial\mathcal{M}} d^4 x \sqrt{-h} K}_{S_{\text{GHY}}} - \underbrace{\sigma \int_{\Sigma} d^4 x \sqrt{-\gamma}}_{S_{\text{brane}}}. \quad (1)$$

Parameters [BL]/[P].

- $\kappa_5^2 = 8\pi G_5$: 5D gravitational coupling (to be determined).
- Λ_5 : bulk cosmological constant (set to zero for simplicity in this analysis).
- σ : brane tension (energy per unit 4-volume, i.e., energy/area in static slicing).

Goal. Determine whether κ_5^2 can be expressed as a function of σ alone:

$$\kappa_5^2 \stackrel{?}{=} f(\sigma). \quad (2)$$

3 Dimensional Analysis

Dimensional Ledger

Natural units ($c = \hbar = 1$):

All quantities expressed in powers of length L .

Quantity	Dimension	Notes
$[\sigma]$	L^{-4}	Energy/area = $M \cdot L^{-2} = L^{-1} \cdot L^{-2} = L^{-4}$
$[G_5]$	L^3	5D Newton constant
$[\kappa_5^2]$	L^3	Since $\kappa_5^2 = 8\pi G_5$
$[\Lambda_5]$	L^{-2}	Cosmological constant
$[K_{\mu\nu}]$	L^{-1}	Extrinsic curvature

Key observation [Dc]. From σ alone, we can construct a characteristic length:

$$\ell_\sigma \equiv \sigma^{-1/4}, \quad [\ell_\sigma] = L. \quad (3)$$

This is the *only* length scale derivable from σ .

Dimensional possibility for κ_5^2 [Dc]. Since $[\kappa_5^2] = L^3$, we need three powers of length. Using ℓ_σ :

$$\kappa_5^2 = C \cdot \ell_\sigma^3 = C \cdot \sigma^{-3/4}, \quad (4)$$

where C is a *dimensionless constant* not determined by dimensional analysis.

Dimensional result: The relation $\kappa_5^2 \propto \sigma^{-3/4}$ is the unique dimensionally consistent possibility if σ is the only input. However, the proportionality constant C is not fixed.

4 Attempt A: Test Consistency with Junction Conditions

Israel junction conditions [BL]. At the brane Σ , the jump in extrinsic curvature satisfies:

$$[K_{\mu\nu}] - [K]\gamma_{\mu\nu} = -\kappa_5^2 \left(S_{\mu\nu} - \frac{1}{3} S \gamma_{\mu\nu} \right). \quad (5)$$

Pure tension brane [BL]. For $S_{\mu\nu} = -\sigma\gamma_{\mu\nu}$ (tension only), with $S = -4\sigma$:

$$[K_{\mu\nu}] - [K]\gamma_{\mu\nu} = -\kappa_5^2 \left(-\sigma + \frac{4\sigma}{3} \right) \gamma_{\mu\nu} = -\frac{\kappa_5^2 \sigma}{3} \gamma_{\mu\nu}. \quad (6)$$

Taking the trace ($\gamma^{\mu\nu}\gamma_{\mu\nu} = 4$):

$$-3[K] = -\frac{4\kappa_5^2 \sigma}{3} \Rightarrow [K] = \frac{4\kappa_5^2 \sigma}{9}. \quad (7)$$

Thus the curvature scale is:

$$k \equiv [K] \sim \kappa_5^2 \sigma. \quad (8)$$

Substitute the dimensional ansatz [Dc]. Using $\kappa_5^2 = C\sigma^{-3/4}$:

$$k \sim C\sigma^{-3/4} \cdot \sigma = C\sigma^{1/4}. \quad (9)$$

The corresponding curvature length is:

$$\ell_k \equiv k^{-1} \sim C^{-1}\sigma^{-1/4} = C^{-1}\ell_\sigma. \quad (10)$$

Consistency check: PASS.

The ansatz $\kappa_5^2 = C\sigma^{-3/4}$ is dimensionally consistent with the Israel junction conditions. The curvature scale $k \sim \sigma^{1/4}$ and length $\ell_k \sim \ell_\sigma$ are well-defined for any $C > 0$.

But: The constant C remains undetermined.

5 Attempt B: Can C Be Fixed by Normalization?

The residual freedom [I]. The action (1) has a normalization freedom:

- Rescaling $g_{AB} \rightarrow \lambda^2 g_{AB}$ changes $\sqrt{-g}$, R , and K .
- Correspondingly, κ_5^2 and σ can be redefined to keep the action invariant.

This is a *coordinate/unit choice*, not physics.

Can EDC fix C ? [OPEN]. For C to be determined, EDC would need to provide one of:

1. A canonical definition of σ in terms of a fundamental action density (with fixed normalization).
2. An independent equation relating κ_5^2 to σ (beyond dimensional consistency).
3. A matching condition to another sector (e.g., EM) that breaks the freedom.

None of these is currently available in EDC.

Underdetermination Statement

Proposition 1 (Underdetermination). *Let σ be the only input parameter with $[\sigma] = L^{-4}$. Then:*

1. *The unique dimensionally consistent relation is $\kappa_5^2 = C \cdot \sigma^{-3/4}$.*
2. *The constant C cannot be fixed by σ alone.*
3. *The junction conditions are satisfied for any $C > 0$, with curvature scale $k = C\sigma^{1/4}$.*

Therefore, σ alone does not determine κ_5^2 uniquely.

6 What Would Fix C ?

Possible additional inputs [OPEN].

Input	Dimension	How it could fix C
Λ_5 (bulk cosm. const.)	L^{-2}	Provides second scale; C from σ/Λ_5^2
ℓ_5 (fundamental 5D length)	L	Direct: $\kappa_5^2 \sim \ell_5^3$; then $C = (\ell_5/\ell_\sigma)^3$
M_5 (5D Planck mass)	L^{-1}	$\kappa_5^2 = M_5^{-3}$; relates to σ via dynamics
Matching to G_N	—	Circular (prohibited)

EDC-internal candidates [I]. If EDC defines σ via a specific microscopic mechanism (e.g., $\sigma = 2\pi R_\xi^2 \rho_P$ for some pressure ρ_P), then additional structure enters. But this is not a σ -only derivation; it imports the scale R_ξ .

7 Dependency Schematic

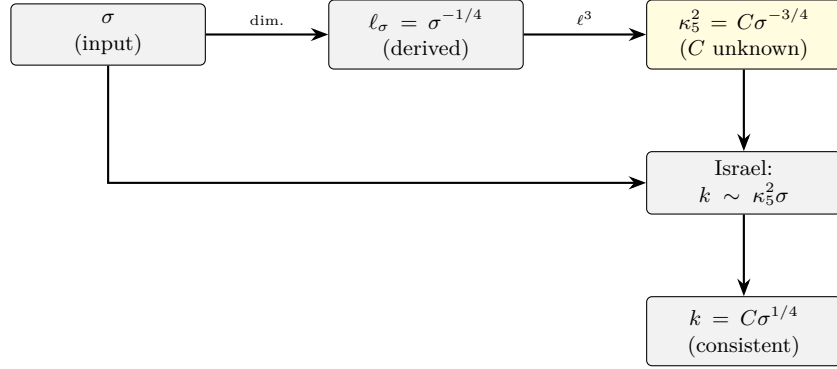


Figure 1: Derivation schematic (program overview; not a result). The constant C remains undetermined by σ alone.

8 Outcome

Underdetermination Statement

Outcome: INCONCLUSIVE.

- **Dimensional analysis:** $\kappa_5^2 = C\sigma^{-3/4}$ is the unique consistent form.
- **Junction conditions:** Satisfied for any $C > 0$.
- **Constant C :** Not fixed by σ alone; requires additional input.
- **No-go:** σ determines ℓ_σ but not κ_5^2 uniquely.

Missing element: An independent scale or normalization condition to fix C .

9 Bridge to BLOCK-003

From derivation_v2, setting $L = R_\xi$ gives:

$$G_N = \frac{\kappa_5^2}{6\pi R_\xi}. \quad (11)$$

Substituting $\kappa_5^2 = C\sigma^{-3/4}$:

$$G_N = \frac{C\sigma^{-3/4}}{6\pi R_\xi}. \quad (12)$$

This still contains the undetermined constant C .

Conclusion: Even combining derivation_v2 ($L = R_\xi$) with derivation_v3 ($\kappa_5^2 \propto \sigma^{-3/4}$), BLOCK-003 remains OPEN because C is not fixed.

The missing link is now narrowed to: **fix** C (or equivalently, provide an independent length scale that sets κ_5^2 absolutely).

10 Conclusion

1. σ provides a natural length scale $\ell_\sigma = \sigma^{-1/4}$.
2. Dimensional analysis gives $\kappa_5^2 = C\sigma^{-3/4}$ as the unique consistent form.
3. The Israel junction conditions are satisfied for any $C > 0$.
4. The constant C cannot be fixed by σ alone.
5. **BLOCK-003 remains OPEN**; the missing element is narrowed to fixing C .

Next step (out of scope). Investigate whether EDC's definition of σ (via membrane geometry or pressure integration) provides the normalization needed to fix C .